related, and it offers rigorous explanations for how and why things work. We could teach RL without any math and just use Python, but that approach would handicap you in understanding future advances.

So we think the math is important, but as our editor noted, there's a common saying in the publishing world: "for every equation in the book, the readership is halved," which probably has some truth to it. There's an unavoidable cognitive overhead in deciphering complex math equations, unless you're a professional mathematician who reads and writes math all day. Faced with wanting to present a rigorous exposition of DRL to give readers a top-rate understanding, and yet wanting to reach as many people as possible, we came up with what we think is a very distinguishing feature of this book. As it turns out, even professional mathematicians are becoming tired of traditional math notation with its huge array of symbols, and within a particular branch of advanced mathematics called *category theory*, mathematicians have developed a purely graphical language called *string diagrams*. String diagrams look very similar to flowcharts and circuit diagrams, and they have a fairly intuitive meaning, but they are just as rigorous and precise as traditional mathematical notations largely based on Greek and Latin symbols.

Figure 1.14 shows a simple example of one type of string diagram that depicts, at a high level, a neural network with two layers. Machine learning (especially deep learning) involves a lot of matrix and vector operations, and string diagrams are particularly well-suited to describing these kinds of operations graphically. String diagrams are also great for describing complex processes because we can describe the process at varying levels of abstraction. The top panel of figure 1.14 shows two rectangles representing the two layers of the neural network, but then we can "zoom in" (look inside the box) on layer 1 to see what it does in more detail, which is shown in the bottom panel of figure 1.14.

Figure 1.14. A string diagram for a two-layer neural network. Reading from left to right, the top string diagram represents a neural network that accepts an input vector of dimension *n*, multiplies it by a matrix of dimensions *n x m*, returning a vector of dimension *m*. Then the nonlinear sigmoid activation function is applied to each element in the m-dimensional vector. This new vector is then fed through the same sequence of steps in layer 2, which produces the final output of the neural network, which is a *k*-dimensional vector.